SCIENCE & TECHNOLOGY • RAS PRELIMS NOTES

General Science & Technology

Everyday science, computers, defence & space, genetics, health, environment and Rajasthan agriculture science

Inside these notes

- Concepts in plain words, then the one-liner facts RAS asks
- Disease, missile, breed and vitamin tables in one place
- Memory hooks and exam traps in every chapter

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Notes made for RAS aspirants - simple language, connected concepts, PYQ-based facts.

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How to use: read a chapter fully once, then revise only the tables, MEMORY HOOKS and EXAM TRAPS. Every fact here is written the way RAS papers ask it.

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StuDy

Everyday Science - Physics & Chemistry Around Us

Science in RAS is not about formulas you solve on paper. It is about the science hiding in your kitchen, your bathroom, the road, and the sky. If you understand why a pressure cooker cooks fast, why a mirage appears on a hot Rajasthan road, and why soap lifts grease, you already know most of this chapter.

Units and Measurements

Every measurement needs a standard. The world uses the **SI system** (International System of Units). RAS asks simple matching: which quantity, which unit.

Quantity	SI Unit	Symbol	Simple meaning
Length	metre	m	distance
Mass	kilogram	kg	amount of matter
Time	second	s	duration
Temperature	kelvin	K	hotness (0 K = absolute zero)
Electric current	ampere	A	flow of charge
Amount of substance	mole	mol	count of particles
Luminous intensity	candela	cd	brightness of light

- **Light year** is a unit of **distance**, not time — the distance light travels in one year (about 9.46 trillion km).
- **Parsec** is the largest commonly used unit of distance in astronomy.
- A **knot** measures the speed of ships and aircraft.
- Sound intensity is measured in **decibel (dB)**; earthquake intensity is discussed on the **Richter scale**.
- **Horsepower** is a unit of power: 1 HP = 746 watts.

MEMORY HOOK: Remember the seven base units as "Me-Ki-Se-Ke-A-Mo-Ca" (metre, kilogram, second, kelvin, ampere, mole, candela).

EXAM TRAP: A light year sounds like time but measures distance. This one-liner is a favourite RAS trap.

Motion and Force - Newton's Laws

Motion means change of position. Force is a push or pull that can change motion. Newton gave three laws, and each one is visible daily.

Newton's three laws with daily examples

- **First Law (law of inertia)**: a body stays at rest or keeps moving in a straight line unless an outside force acts. Example: when a bus suddenly starts, passengers fall backward; when it brakes, they fall forward. **Inertia** depends on mass — a heavier body has more inertia.
- **Second Law**: force = mass x acceleration (F = ma). Example: it takes more force to push a loaded truck than an empty cart. This law defines the unit of force, the **newton**.
- **Third Law**: every action has an equal and opposite reaction. Example: a rocket throws gases downward and is pushed upward; a swimmer pushes water back and moves forward.

- **Friction** opposes motion between surfaces. It lets us walk and lets brakes work, but it also causes wear. Ball bearings reduce friction in machines.
- A body moving in a circle needs a **centripetal force** towards the centre. A cyclist leans inward while turning for this balance.
- **Momentum** = mass x velocity. It explains why a speeding truck is harder to stop than a cycle at the same speed.

PYQ PATTERN: RAS often asks for the daily-life example of the third law — rocket propulsion and swimming are the standard answers.

Work, Energy and Power

- **Work** is done when a force moves an object in its direction. Work = force x distance. Unit: **joule**.
- **Energy** is the capacity to do work. Unit: also **joule**. One calorie = 4.186 joules.
- **Kinetic energy** is energy of motion; **potential energy** is energy stored due to position (water stored in a dam).
- **Law of conservation of energy**: energy can neither be created nor destroyed, only converted — a bulb converts electrical energy into light and heat.
- **Power** is the rate of doing work. Unit: **watt**. Your electricity bill counts energy in **kilowatt-hour (kWh)**, popularly called one "unit".

EXAM TRAP: The electricity "unit" on a bill is a unit of energy (kWh), not of power.

Energy conversions in everyday devices

Device	Energy conversion
Electric bulb / LED	electrical to light (and some heat)
Electric motor	electrical to mechanical
Dynamo / generator	mechanical to electrical
Solar cell	light (solar) to electrical
Battery (while discharging)	chemical to electrical
Loudspeaker	electrical to sound
Photosynthesis	light to chemical (food) energy in plants

- **Renewable sources** (solar, wind, hydro, biomass) refill naturally; **non-renewable sources** (coal, petroleum, natural gas) finish once used. Rajasthan's huge solar parks in the Thar (like the Bhadla Solar Park near Jodhpur, among the world's largest) turn the desert's biggest problem — fierce sunshine — into its biggest resource.

MEMORY HOOK: Motor and dynamo are mirror images: motor eats electricity and gives motion; dynamo eats motion and gives electricity. Options swap them on purpose.

Heat and Temperature

Temperature tells how hot a body is; **heat** is energy in transfer. Three scales are used.

Scale	Water freezes	Water boils	Used where
Celsius	0 degree	100 degree	daily life, India
Fahrenheit	32 degree	212 degree	USA, body thermometers

Scale	Water freezes	Water boils	Used where
Kelvin	273 K	373 K	science; begins at absolute zero

- Conversion: C/5 = (F − 32)/9; K = C + 273.
- At **−40 degrees**, Celsius and Fahrenheit are equal — a neat one-liner fact.
- A clinical thermometer reads about 35 to 42 degree C; normal body temperature is about **37 degree C (98.6 degree F)**.
- Water has an unusually **high specific heat**, so it heats and cools slowly — this is why coastal places have mild weather and why water is used as a coolant in engines.
- A **pressure cooker** cooks faster because higher pressure raises the boiling point of water above 100 degree C, so food gets hotter steam/water contact.

MEMORY HOOK: Pressure high, boiling point high — that is the whole pressure-cooker secret.

Light - Mirrors, Lenses and Daily Phenomena

Light travels in straight lines and reflects and refracts (bends) when it changes medium.

Mirrors and lenses in daily use

Device	Type used	Why
Vehicle rear-view mirror	convex mirror	wide field of view, erect smaller image
Dentist's mirror	concave mirror	magnified view of teeth
Shaving mirror	concave mirror	magnified face image
Torch / headlight reflector	concave mirror	collects light into a beam
Camera & spectacles	convex/concave lenses	correction of vision defects
Simple microscope	convex lens	magnifies small objects

- **Myopia (short sight)**: person sees near objects clearly, distant ones blur; corrected by a **concave (diverging) lens**.
- **Hypermetropia (long sight)**: distant objects clear, near ones blur; corrected by a **convex (converging) lens**.
- **Presbyopia** in old age is corrected with **bifocal lenses**.
- The power of a lens is measured in **dioptre**. The human eye lens changes shape (accommodation) to focus.

Total Internal Reflection (TIR)

When light travelling in a denser medium (like water or glass) hits the boundary at a large angle, it reflects completely back inside instead of passing out. This is **total internal reflection**. Conditions: light must go from denser to rarer medium, and the angle must exceed the **critical angle**.

- **Mirage**: on a hot road in Rajasthan, air near the ground becomes very hot and rarer. Light from the sky bends and finally reflects totally, so the traveller sees what looks like water on the road. It is an optical illusion caused by TIR.
- **Optical fibres** work on TIR — light signals bounce inside a thin glass fibre with almost no loss. Used in internet cables and endoscopy.
- The **brilliance of diamond** is also due to TIR inside its cut faces.

PYQ PATTERN: Mirage = total internal reflection is one of the most repeated facts in Rajasthan exams. Lock it in.

Sound Basics

- Sound is a **longitudinal mechanical wave** — it needs a medium (solid, liquid, gas) and cannot travel in vacuum.
- Speed of sound in air at room temperature is about **343 m/s**. Sound travels fastest in solids, slower in liquids, slowest in gases.
- Human hearing range: **20 Hz to 20,000 Hz**. Below 20 Hz is **infrasonic** (elephants, earthquakes); above 20,000 Hz is **ultrasonic** (bats, SONAR, ultrasound scanning in medicine).
- **Echo** is reflected sound; the reflecting surface must be at least about 17 metres away for a clear echo.
- **SONAR** uses ultrasound to measure sea depth and locate objects underwater.

EXAM TRAP: Light can travel in vacuum (it comes from the Sun), sound cannot. Options often swap this.

Electricity and Pressure in Daily Life

- **Electric current** is the flow of electrons. **Direct current (DC)** flows one way (batteries, solar cells); **alternating current (AC)** keeps reversing direction and is what comes to our homes (in India at 220-240 volts, 50 Hz).
- A **fuse** is a safety wire with a low melting point: it melts and breaks the circuit when current becomes dangerously high. The modern replacement is the **MCB (Miniature Circuit Breaker)**, which trips automatically.
- **Earthing** gives leaked current a safe path into the ground — that is why the third pin of a plug can save a life.
- Copper is used for wiring because it is a very good conductor with low resistance; thick wires are used for heavy appliances because they heat up less.
- **Pressure** = force per unit area. A sharp knife cuts better because the same force acts on a tiny area, giving huge pressure.
- The **hydraulic lift and hydraulic brakes** in vehicles use Pascal's law: pressure applied to a liquid spreads equally in all directions, so a small force on a small piston creates a large force on a bigger piston.
- Blood pressure is measured with a **sphygmomanometer**; air pressure with a **barometer**.

MEMORY HOOK: Fuse melts, MCB trips, earth wire saves — three separate safety ideas that options love to mix up.

Household Chemistry - Acids, Bases and pH

An **acid** turns blue litmus red and tastes sour (curd, lemon, vinegar). A **base** turns red litmus blue and feels soapy (soap solution, washing soda). The **pH scale** (0 to 14) tells strength: below 7 acidic, 7 neutral, above 7 alkaline/basic.

- Our stomach produces **hydrochloric acid (HCl)** to digest food. Excess acid causes acidity, cured by mild bases called **antacids** like magnesium hydroxide (milk of magnesia).
- Bee sting injects an acid (formic/methanoic acid); rubbing a mild base like baking soda gives relief. A wasp sting is alkaline and is treated with a mild acid like vinegar.

- Tooth decay starts when mouth pH falls below about **5.5**; toothpaste is basic and neutralises acid.
- Acid rain (pH below about 5.6) harms crops, buildings and the Taj Mahal's marble.
- Farmers treat acidic soil with **quicklime or slaked lime**, and basic soil with organic matter/gypsum as needed.

MEMORY HOOK: Bee = Acid sting → Base cures. Wasp = Alkaline sting → Acid (vinegar) cures. "B-A, W-A" — the first letters pair up.

Soaps and Detergents

- **Soap** is made by **saponification** — heating fats/oils with an alkali like NaOH. Glycerol is a useful by-product.
- Soap cleans because one end of its molecule loves water and the other end loves grease; the grease end traps dirt in tiny balls called **micelles**, which water washes away.
- Soap does **not** work well in **hard water** (water with calcium and magnesium salts) — it forms sticky scum.
- **Detergents** work in hard water too, so they replaced soaps for washing clothes. Detergent pollution, however, can foam in water bodies.

Common Chemicals Table

Common name	Chemical name / formula	Everyday use
Washing soda	Sodium carbonate, Na2CO3.10H2O	softening water, making glass, washing
Baking soda	Sodium bicarbonate, NaHCO3	cooking, antacid, fire extinguishers
Caustic soda	Sodium hydroxide, NaOH	soap making, paper
Plaster of Paris	Calcium sulphate hemihydrate, CaSO4.1/2H2O	casts for fractures, statues, ceilings
Bleaching powder	Calcium oxychloride, CaOCl2	disinfecting water, bleaching
Gypsum	Calcium sulphate dihydrate, CaSO4.2H2O	making plaster of Paris, cement
Common salt	Sodium chloride, NaCl	food, preservation
Slaked lime	Calcium hydroxide, Ca(OH)2	whitewashing, soil treatment
Quicklime	Calcium oxide, CaO	cement, drying agent

- Adding water to **Plaster of Paris** sets it into hard gypsum — that is why fracture casts harden.
- **Bleaching powder** releases chlorine, which kills germs in drinking water.

EXAM TRAP: Washing soda (Na2CO3) vs baking soda (NaHCO3) is a classic confusion pair in RAS papers.

Metals and Non-Metals - Everyday Facts

- **Mercury** is the only metal liquid at room temperature; **bromine** is the only non-metal liquid at room temperature.
- **Tungsten** is used for bulb filaments because its melting point is very high. **Copper and aluminium** are used for wires because they conduct electricity well.
- **Gold and platinum** do not rust or tarnish — called noble metals.

- Iron rusts in moist air forming **rust (hydrated iron oxide)**. Painting, oiling and **galvanisation** (coating iron with zinc) prevent rusting.
- **Stainless steel** is an alloy of iron with chromium and nickel. Other alloys: **brass** (copper + zinc), **bronze** (copper + tin), **solder** (lead + tin).
- Aluminium is light and corrosion-resistant, so it is used in aircraft bodies and cooking foil — even though it is a reactive metal, its thin oxide layer protects it.

PYQ PATTERN: Alloy composition questions (brass, bronze, stainless steel) appear again and again. Learn the three pairs cold.

Computers & ICT

A computer is a machine that takes data (input), processes it by stored instructions, and gives information (output). ICT — Information and Communication Technology — is the larger family: computers, the internet, phones and the networks that connect them.

Computer Generations

Generation	Period (approx.)	Main technology	Example
First	1940s-1950s	Vacuum tubes	ENIAC, UNIVAC
Second	1950s-1960s	Transistors	IBM 1401
Third	1960s-1970s	Integrated circuits (IC)	IBM 360
Fourth	1970s onwards	Microprocessor (VLSI)	Personal computers
Fifth	Present/future	Artificial intelligence, ULSI	AI systems, quantum research

- **Charles Babbage** is called the father of the computer for his Analytical Engine design; **Ada Lovelace** is regarded as the first programmer.
- India's first computer, **TIFRAC**, was built at the Tata Institute of Fundamental Research, Mumbai (1960).

MEMORY HOOK: Generation key = the switching device: Tubes → Transistors → ICs → Microprocessor → AI. RAS asks the device, not the dates.

Hardware and Software

Device type	Examples	Direction of data
Input devices	keyboard, mouse, scanner, microphone, webcam	send data into the computer
Output devices	monitor, printer, speakers, projector	give results out to the user
Input + output	touchscreen, pen drive (storage read/write)	both ways

- **Hardware** = the physical parts you can touch: CPU, monitor, keyboard, printer.
- **Software** = the programs and instructions. Two broad types: **system software** (operating systems like Windows, Linux, Android) and **application software** (Word, browsers, games).
- The **CPU (Central Processing Unit)** is the brain; it has the ALU (arithmetic and logic) and the Control Unit.

- **RAM** is temporary working memory — data is lost when power is off (volatile). **ROM** keeps start-up instructions permanently (non-volatile).
- Storage devices: hard disk, SSD, pen drives, memory cards, CDs/DVDs.

Memory Units

Computers count in binary — 0 and 1. One binary digit is a **bit**; 8 bits make a **byte**.

Unit	Size
1 byte	8 bits
1 Kilobyte (KB)	1024 bytes
1 Megabyte (MB)	1024 KB
1 Gigabyte (GB)	1024 MB
1 Terabyte (TB)	1024 GB
1 Petabyte (PB)	1024 TB

EXAM TRAP: Computer memory multiplies by **1024**, not 1000. This tiny fact is asked directly.

Operating Systems and Languages

- An **operating system** manages hardware and lets other programs run: Windows, Linux, macOS, Android, iOS.
- Computer languages: machine language (binary) is understood directly by the computer; **high-level languages** like C, C++, Java and Python are human-friendly and need a compiler or interpreter to convert them.
- A **compiler** converts the whole program at once; an **interpreter** converts it line by line.

The Internet - How It Works in Simple Words

The internet is a worldwide network of networks. Your device connects to an **ISP (Internet Service Provider)**; data travels in small packets through routers and undersea cables to servers anywhere on Earth and back in fractions of a second.

- **WWW (World Wide Web)** is not the same as the internet. The Web is one service (pages linked together, invented by **Tim Berners-Lee**) that runs **on** the internet. Email and chats are other services using the same internet.
- **Website address (URL)** names a page; the **Domain Name System (DNS)** converts names like a site address into numeric IP addresses.
- **Search engines** (Google, Bing) index billions of pages and find matching ones for your keywords.
- **Email** sends messages between mail servers using protocols like SMTP (sending) and IMAP/POP3 (receiving).
- **Wi-Fi** connects devices wirelessly over short distances; mobile data (4G/5G) connects through towers over long distances.

EXAM TRAP: "Internet and WWW are the same thing" — false. WWW is a service that uses the internet. Another trap: Google is a search engine, not the internet itself.

Key Terms Table

Term	Meaning in simple words
AI (Artificial Intelligence)	machines doing tasks that need human-like thinking
Machine Learning	AI that learns patterns from data instead of fixed rules
IoT (Internet of Things)	everyday devices (fans, meters, tractors) connected to the internet
Cloud Computing	renting computing and storage over the internet instead of owning machines
Big Data	datasets so huge that special tools are needed to analyse them
Blockchain	a shared, tamper-resistant digital ledger; records linked in blocks
5G	fifth-generation mobile network — faster, lower delay, many devices
QR code	a square printed code a phone camera reads instantly
UPI	Unified Payments Interface — instant bank-to-bank payment through apps
GPS	satellite-based system that tells your location

Cyber Security Basics and Digital India

- Common threats: **virus** (harmful program), **phishing** (fake messages/pages that steal passwords or OTPs), **ransomware** (locks your files and demands money).
- Safety rules: never share OTPs or PINs, use strong passwords, turn on two-factor authentication, and check the sender before clicking links.
- **Digital India** (launched 2015) is the Government of India's flagship programme with three pillars: digital infrastructure, governance and services on demand, and digital empowerment of citizens.
- India's digital public systems — **Aadhaar, UPI, DigiLocker, CoWIN** — are studied worldwide as examples of large-scale public digital platforms.

PYQ PATTERN: Questions on UPI, Digital India pillars and phishing appear regularly now; treat ICT as a scoring area, not a side topic.

ICT in Daily Governance

- **e-Governance** means delivering government services through digital channels: certificates, land records and bill payment online instead of office queues. Rajasthan's **e-Mitra** kiosks are the state's well-known ground-level model of this idea, and **Jan Aadhaar** gives families a single identity for scheme benefits.
- **Digital payment rails** — UPI, IMPS and Aadhaar-enabled payments — let even a street vendor receive money instantly without cash or a card machine.
- Satellite-based services complete the ICT picture: telemedicine links village patients to city doctors, and tele-education beams lessons to remote schools.

PYQ PATTERN: e-Mitra (Rajasthan) and Jan Aadhaar are the two state-level ICT names most likely to appear in a Rajasthan paper.

Supercomputers of India

A supercomputer does trillions of calculations per second for weather, defence, research and AI.

- The **PARAM series**, developed by **C-DAC (Centre for Development of Advanced Computing), Pune**, began with **PARAM 8000 (1991)** — a landmark of Indian indigenisation when export of supercomputers to India was restricted.
- **Pratyush** (at IITM, Pune) and **Mihir** (at NCMRWF, Noida) are high-performance systems used mainly for **weather and climate forecasting** — together they serve monsoon prediction, cyclone warnings and ocean state forecasts.
- Newer national systems under the **National Supercomputing Mission** include PARAM Siddhi-AI and AIRAWAT for AI research.

MEMORY HOOK: Weather twins — Pratyush (Pune) and Mihir (Noida) work for the monsoon. PARAM belongs to C-DAC, Pune.

Defence & Space Technology of India

India's defence and space programmes are twins of self-reliance: both began under technology denial and grew by building at home. RAS asks names, types and firsts — a few clean tables will earn easy marks here.

DRDO in Brief

- **DRDO (Defence Research and Development Organisation)** was formed in **1958**. It designs weapons, missiles, radars, aircraft systems and life-support technology for the armed forces, working through a network of laboratories across India.
- DRDO's famous product family is missiles, but it also works on electronic warfare, unmanned vehicles, naval systems and soldier support (like high-altitude clothing and food).

The Missile Family

The **Integrated Guided Missile Development Programme (IGMDP)**, started in **1983** under the leadership of **Dr A.P.J. Abdul Kalam** (called the "Missile Man of India"), developed five core missiles: Prithvi, Agni, Akash, Trishul and Nag. Remember them as **"PAATN"**.

Missile	Type	Key point
Prithvi	surface-to-surface, short range	first missile of the IGMDP; liquid fuel
Agni I-V	surface-to-surface, medium to intercontinental range	strategic (nuclear-capable) ballistic series; Agni-V range over 5,000 km
Akash	surface-to-air, medium range	hits enemy aircraft and drones
BrahMos	supersonic cruise missile	joint venture of India and Russia; among the fastest cruise missiles; can be launched from land, ship, submarine and aircraft
Nag	anti-tank guided missile	"fire and forget" — locks on the tank and needs no further guidance
Trishul	surface-to-air, short range	legacy system of the IGMDP; programme closed after technology demonstration
Prahaar	surface-to-surface tactical missile	short-range battlefield support weapon

- Ballistic missiles fly in a high arc like a thrown stone and are powered only in the first phase; **cruise missiles** fly low, powered throughout, like a pilotless aircraft guided to the target.
- **BrahMos** takes its name from the **Brahmaputra** and **Moskva** rivers — a favourite one-liner question.

EXAM TRAP: BrahMos is a **cruise** missile, not ballistic, and it is **supersonic**, not subsonic. Also, it is jointly developed with Russia, not a fully indigenous-only project.

PYQ PATTERN: The IGMDP five (Prithvi, Agni, Akash, Trishul, Nag) and Kalam's role are asked repeatedly in RAS pre papers.

Aircraft and Ships

Platform	What it is	Key point
Tejas	Light Combat Aircraft (LCA)	India's indigenous single-engine fighter; designed by ADA, built by HAL
INS Vikrant	Aircraft carrier	India's first aircraft carrier built in India (at Cochin Shipyard); commissioned 2022
Arihant class	Nuclear-powered submarines	give India the sea leg of its nuclear triad

India's Space Journey - ISRO

- **ISRO (Indian Space Research Organisation)** was formed in **1969** at Bengaluru. Its father figure is **Dr Vikram Sarabhai**, founder of the Indian space programme.
- Rockets are launched mainly from **Sriharikota (Satish Dhawan Space Centre)** in Andhra Pradesh, on the east coast, so debris falls into the sea and Earth's rotation gives an eastward boost.

Launch vehicles

Vehicle	Nickname/strength	Used for
PSLV	the "workhorse of ISRO"	medium payloads; earth observation and many foreign satellites; also launched Mangalyaan and Chandrayaan-1
GSLV	heavier lifter with cryogenic upper stage	heavier satellites to higher orbits
LVM3 (earlier called GSLV Mk III)	heaviest Indian rocket	Chandrayaan-2 and Chandrayaan-3; chosen for Gaganyaan human flights

Milestones table

Mission	Year	Achievement
Aryabhata	1975	India's first satellite (launched with Soviet help)
Chandrayaan-1	2008	India's first Moon mission; its instruments confirmed **water molecules on the Moon**

Mission	Year	Achievement
Mangalyaan (Mars Orbiter Mission)	2013	India became the **first country to reach Mars orbit in the first attempt**; launched by PSLV
Chandrayaan-3	2023	First-ever soft landing near the **lunar south pole** (23 August 2023); lander Vikram, rover Pragyan; the day is now **National Space Day**
Aditya-L1	2023	India's first solar observatory, placed in a halo orbit around the **L1 point** to study the Sun continuously
NavIC	operational regional system	India's own satellite navigation system covering India and nearby region
GSAT series	ongoing	communication satellites for TV, telephony, broadband and weather data links

- India's satellites fall into working families. The **INSAT/GSAT series** handles communication, TV and weather watch from high geostationary orbit (about 36,000 km up, so the satellite appears fixed over one place). The **IRS/Cartosat/RISAT family** does remote sensing — photographing Earth for crop estimation, water mapping, forest survey, disaster watch and defence. **IRNSS/NavIC** gives navigation.
- Remote sensing is an everyday tool in Rajasthan: satellite images help locate groundwater, track desert spread, plan watershed works and estimate crops before harvest.

Orbits in simple words

- **Low Earth Orbit (LEO)**, a few hundred kilometres up: best for sharp Earth photographs, so remote-sensing satellites and space stations live here. Gaganyaan will also fly in LEO.
- **Geostationary Orbit (GEO)**, about 36,000 km above the equator: the satellite circles at exactly Earth's rotation speed, so it hangs over the same spot forever — perfect for TV broadcast and weather watch (INSAT/GSAT).
- **Polar/sun-synchronous orbits** pass over the poles while Earth rotates below, letting one satellite photograph the whole planet strip by strip — the standard path for IRS remote-sensing satellites.
- The **L1 point** (Lagrange point 1), about 1.5 million km towards the Sun, lets a spacecraft watch the Sun without Earth's shadow interrupting — that is where Aditya-L1 sits.

MEMORY HOOK: Near orbit for cameras, far fixed orbit for TV — height decides the job.

- **Gaganyaan** is India's human spaceflight programme: Indian astronauts (Gaganyatris) will travel in an Indian crew module on LVM3 to low Earth orbit. Uncrewed test missions come first.
- The **cryogenic engine** technology (using super-cold liquid hydrogen and oxygen) was a famous indigenisation success — India developed it after being denied the technology from abroad.

EXAM TRAP: Mangalyaan flew on **PSLV**, not GSLV. And Chandrayaan-3 landed near the Moon's **south pole** region — no country had soft-landed there before.

MEMORY HOOK: "PSLV went to Mars" sounds wrong but is exactly right — RAS loves testing that surprise.

Inheritance, Variation & Genetic Engineering

Children resemble parents but are never exact copies. **Heredity** is the passing of traits from parents to offspring; **variation** is the small differences that make each individual unique. This chapter covers the rules of heredity and how humans now edit those rules deliberately.

Mendel's Laws

Gregor Johann Mendel, an Austrian monk, experimented with **garden pea plants** in the 1860s and is called the **father of genetics**. He studied contrasting traits like tall/dwarf plants and round/wrinkled seeds.

- **Unit of heredity**: the **gene** — a segment of **DNA** that controls a trait. Different forms of a gene are called **alleles** (e.g., tall allele T, dwarf allele t).
- **Law of Dominance**: in a mixed pair (Tt), only the **dominant** trait shows; the hidden one is **recessive** and reappears in the next generation in a 3:1 ratio (monohybrid cross).
- **Law of Segregation**: the two alleles of a pair separate during gamete formation, so each gamete carries only one allele.
- **Law of Independent Assortment**: genes of different traits are inherited independently (dihybrid cross gives a 9:3:3:1 ratio).

MEMORY HOOK: Ratios to lock: 3:1 (one trait, second generation) and 9:3:3:1 (two traits). Mendel = pea plant = father of genetics.

DNA and RNA

Feature	DNA	RNA
Full name	Deoxyribonucleic acid	Ribonucleic acid
Sugar	deoxyribose	ribose
Shape	double helix (two strands)	single strand (usually)
Bases	A, T, G, C	A, U, G, C (uracil replaces thymine)
Job	stores heredity information	carries instructions to build proteins

- The double helix model was given by **Watson and Crick (1953)**.
- DNA is packaged as **chromosomes** inside the nucleus. Humans have **46 chromosomes (23 pairs)** in body cells; gametes (sperm/egg) carry 23.
- Genes make proteins, and proteins build traits — eye colour, blood group, enzymes, everything.

Sex Determination and Blood Groups

- Of the 23 pairs, one pair is the **sex chromosomes**: XX in females, XY in males. The father's sperm (carrying X or Y) decides the baby's sex — biologically, sex is determined by the male gamete.
- **ABO blood groups** are controlled by one gene with three alleles (A, B, O). A and B are co-dominant (both show in AB group); O is recessive.
- The **Rh factor** is another protein on red blood cells. An Rh-negative mother carrying an Rh-positive baby can face problems in later pregnancies (erythroblastosis fetalis) — prevented by an injection after the first delivery.

Blood group	Can receive from	Note
O negative	O negative only	universal donor (O group broadly)
AB positive	all groups	universal recipient (AB group broadly)

EXAM TRAP: Universal donor is **O**, universal recipient is **AB**. Options frequently interchange them.

Common Genetic Disorders

Disorder	Cause in simple words	Key point
Colour blindness	recessive gene on X chromosome	cannot distinguish red-green; mostly males affected
Haemophilia	recessive gene on X chromosome	blood does not clot; called the "royal disease"
Sickle-cell anaemia	defective haemoglobin gene	red cells become sickle-shaped; common in some tribal belts of India
Thalassemia	defective haemoglobin production	needs regular blood transfusion
Down syndrome	extra chromosome 21 (total 47)	caused by non-disjunction, not by a faulty gene alone

Genetic Engineering

Genetic engineering means deliberately cutting and joining DNA to move a gene from one organism into another. The joined DNA is called **recombinant DNA**. In simple steps: (1) cut the needed gene using "molecular scissors" (restriction enzymes), (2) join it into a carrier (plasmid of a bacterium) using "molecular glue" (ligase), (3) insert it into host cells, (4) grow the cells so each carries and expresses the new gene.

- **Human insulin from bacteria**: the human insulin gene was inserted into E. coli bacteria, which then produce pure insulin for diabetic patients — the first famous product of this technology.
- **Bt cotton**: a gene from the bacterium *Bacillus thuringiensis* (Bt) makes cotton produce a protein toxic to bollworm insects, reducing pesticide sprays. Bt brinjal was developed too but was placed under moratorium in India.
- **Golden rice** is engineered to make vitamin A precursor (beta-carotene) to fight vitamin A deficiency.

The Human Genome Project

A **genome** is the complete set of DNA instructions of an organism. The international **Human Genome Project**, completed in **2003**, read the entire human genetic code — about 3 billion base pairs — and mapped the roughly 20,000-25,000 human genes. Its legacy is personalised medicine: reading a patient's genes to predict disease risk (like some cancers) and choose better treatment. India contributes through its own genomics research institutes.

MEMORY HOOK: 46 chromosomes, about 3 billion letters — the genome is the instruction book of a human being.

Biotechnology Applications

- **IVF (In Vitro Fertilisation)**: fertilisation happens in a laboratory dish and the early embryo is placed in the mother's womb — the "test-tube baby" technique. India's first test-tube baby was born in 1978, the same year as the world's first (Louise Brown, England).
- **Cloning**: making a genetically identical copy of an organism. **Dolly the sheep (1996, Scotland)** was the first mammal cloned from an adult body cell.
- **Stem cells**: master cells that can develop into many cell types. They hold promise for repairing damaged tissues; embryonic and adult (like bone marrow) stem cells are the two broad types.
- **DNA fingerprinting**: every person's DNA pattern (except identical twins) is unique. Developed by **Alec Jeffreys**, it is used in crime investigation, disaster identification and paternity disputes. In India, this field was pioneered by **Dr Lalji Singh** (CCMB, Hyderabad).

PYQ PATTERN: Expect one-liners: Dolly = first cloned mammal, insulin = first recombinant product, Lalji Singh = father of DNA fingerprinting in India.

Nanotechnology

"Nano" is about size. A **nanometre (nm)** is one-billionth of a metre — about 100,000 times thinner than a human hair. **Nanotechnology** works with matter at roughly **1 to 100 nm**, the scale of atoms and molecules.

Why Properties Change at Nano Scale

At very small sizes, most atoms of a substance sit on the **surface** rather than inside, so reactions become much faster. Also, quantum effects appear: colours, strength, melting point and electrical behaviour can change completely. Nano gold, for example, can look red or purple instead of yellow, and nano particles can slip into cells that normal particles cannot enter.

MEMORY HOOK: Same substance, new behaviour — only because the pieces are tiny. That is the whole idea of nano.

Applications

Field	Use
Medicine	targeted drug delivery to cancer cells, nano diagnostics
Electronics	smaller, faster chips; better displays and memory
Cosmetics	nano zinc oxide and titanium dioxide in sunscreens — transparent sun protection
Water purification	nano filters and nano silver to kill germs and remove impurities
Textiles & sport	stain-resistant fabrics, stronger and lighter tennis rackets and bats
Agriculture	nano fertilisers and pesticides that work in smaller doses

Carbon Nanotubes and Graphene

- **Carbon nanotubes** are rolled sheets of carbon atoms — extremely strong and light, excellent conductors; used in composites and electronics research.

- **Graphene** is a single layer of carbon atoms arranged in a honeycomb — among the strongest and thinnest materials known, and a superb conductor. Its isolation won the 2010 Nobel Prize in Physics.

India's Nano Mission

India launched the **Nano Mission in 2007** under the Department of Science and Technology, with **Prof. C.N.R. Rao** as its leading figure, to build research capacity, centres and trained manpower in nanoscience. Bengaluru hosts a major centre (CeNS and the Institute for Nano Science initiatives). India ranks among the top countries in nano research publications.

EXAM TRAP: Nano Mission year is **2007**, and its scale is 1-100 nm — RAS asks both numbers directly.

Concerns: nano particles are so small that their long-term effects on lungs, skin, soil and water life are not fully known. Safe disposal, labelling and regulation are still catching up — the benefits are real, but caution is necessary.

Nano Around You Already

You may already use nanotechnology daily without knowing it. The anti-glare and scratch-resistant coatings on spectacles and phone screens, the silver particles in some water purifiers and wound dressings, and the invisible zinc oxide in modern sunscreens all work at nano scale. Even computer processors are built with features measured in a few nanometres, which is why each new chip generation packs more power into the same small space.

MEMORY HOOK: If the advertisement says "nano silver purifier" or "5 nm chip", you now know both are talking about the same tiny scale — billionths of a metre.

Food, Nutrition, Health & Diseases

Food is fuel, building material and medicine at once. This chapter links nutrients to deficiency diseases, and infections to the national programmes that fight them — both rich scoring areas in RAS, including Rajasthan's own health picture.

Nutrients - Macro and Micro

Macronutrients are needed in large amounts: **carbohydrates** (energy), **proteins** (body building), and **fats** (concentrated energy, insulation). **Micronutrients** are needed in tiny amounts but are vital: **vitamins** and **minerals**. Water and roughage (fibre) complete the diet — fibre adds bulk and helps bowel movement.

Nutrient	Good sources	Main job	Deficiency effect
Carbohydrates	cereals, potato, sugar	quick energy	weakness, underweight
Proteins	pulses, milk, egg, soyabean	growth and repair	kwashiorkor, marasmus in children
Fats	oils, ghee, nuts	stored energy	energy shortage, vitamin absorption falls
Iron	jaggery, green leafy vegetables, dates	makes haemoglobin	anaemia (very common in Rajasthan women)
Iodine	iodised salt	thyroid function	goitre; mental retardation in severe cases

Nutrient	Good sources	Main job	Deficiency effect
Calcium	milk, ragi, sesame	bones and teeth	weak bones, osteoporosis

- **Kwashiorkor**: protein deficiency — child with swollen belly and thin limbs. **Marasmus**: total food (energy + protein) deficiency — child looks like skin and bones.
- A **balanced diet** contains all nutrients in right proportions for age and work. Rajasthan's traditional bajra, moth dal, curd and seasonal greens form a naturally balanced pattern.

The Vitamin Table - Learn It Cold

Vitamin	Rich sources	Deficiency disease / symptom
A	carrot, papaya, liver, green leafy veg	night blindness, xerophthalmia
B1 (thiamine)	whole grains, legumes	beriberi
B3 (niacin)	grains, groundnut, meat	pellagra (diarrhoea, dermatitis, dementia)
B12	milk, egg, meat (animal foods)	pernicious anaemia
C	amla, guava, citrus fruits	scurvy (bleeding gums)
D	sunlight on skin, fish, fortified foods	rickets in children, osteomalacia in adults
E	vegetable oils, nuts, seeds	nerve/muscle problems (rare)
K	green leafy vegetables, gut bacteria	delayed blood clotting, bleeding

MEMORY HOOK: D is the "sunshine vitamin"; C guards the gums (scurvy hit sailors without citrus); A guards the eyes at night.

EXAM TRAP: Rickets = vitamin D in children; beriberi = B1; scurvy = C. Examiners rotate these three relentlessly.

Diseases - Tables by Agent

Viral diseases

Disease	Agent	Spread	Key point
Dengue	dengue virus	Aedes mosquito bite	platelet fall; mosquito breeds in clean stored water
Hepatitis	hepatitis viruses (A-E)	A & E via contaminated water/food; B & C via blood	jaundice; vaccine exists for B
Polio	poliovirus	faecal-oral route	India declared **polio-free in 2014**
Rabies	rabies virus	bite of infected dog/animal	almost always fatal after symptoms; vaccine + washing the wound immediately saves life
AIDS	HIV virus	blood, unprotected sex, infected needles, mother to child	attacks immune system (CD4 cells); no complete cure, controlled by ART medicines

Disease	Agent	Spread	Key point
COVID-19	SARS-CoV-2 coronavirus	droplets and contact	global pandemic declared 2020; vaccines Covishield and Covaxin used in India

Bacterial diseases

Disease	Agent	Spread	Key point
Tuberculosis (TB)	Mycobacterium tuberculosis	air (cough droplets)	BCG vaccine at birth; treated by DOTS course
Cholera	Vibrio cholerae	contaminated water	severe watery diarrhoea; ORS is the life-saver
Typhoid	Salmonella typhi	contaminated food/water	Widal test used in diagnosis
Plague	Yersinia pestis	rat fleas	the Surat outbreak (1994) is the famous Indian example
Leprosy	Mycobacterium leprae	prolonged contact	curable with MDT medicines

Protozoal, fungal and lifestyle diseases

Disease	Agent / cause	Key point
Malaria	Plasmodium parasite, spread by female **Anopheles** mosquito	fever with chills in cycles; chloroquine and ACT medicines used
Kala-azar	Leishmania parasite, spread by sandfly	"black fever"; endemic in Bihar and eastern India
Ringworm, athlete's foot	fungi	skin infections, spread by contact
Diabetes	insulin shortage or resistance	lifestyle disease; high blood sugar damages organs slowly
Hypertension	persistently high blood pressure	called a "silent killer"; salt, stress, obesity raise risk

EXAM TRAP: Malaria is caused by a **protozoan (Plasmodium)**, spread by a mosquito — the mosquito is the vector, not the cause. Dengue's vector is Aedes; malaria's is Anopheles; kala-azar's is the sandfly. Vector-matching is a RAS favourite.

Immunity and Vaccination

- **Immunity** is the body's defence system, mainly white blood cells and antibodies. **Active immunity** develops when the body itself makes antibodies (after disease or vaccine) and lasts long; **passive immunity** comes ready-made (mother's antibodies to a baby, or injections) and lasts short.
- A **vaccine** is a killed/weakened germ or its part that trains the immune system safely. **Edward Jenner** made the first vaccine (smallpox, 1796); India eradicated smallpox (1980 world eradication) and polio.
- BCG protects against TB, OPV against polio, MMR against measles-mumps-rubella, and hepatitis B vaccine is given at birth under national schedules.

Public Health Programmes

Programme	Launched	Purpose
National Health Mission (NHM)	2013 (merging NRHM 2005 + urban mission)	strengthens public health system, rural focus
Ayushman Bharat - PM-JAY	2018	free hospital cover of Rs 5 lakh per family per year for poor families
Universal Immunisation Programme (UIP)	1985	free vaccines against major childhood diseases
Mission Indradhanush	2014	intensive drive to fully immunise every child and pregnant woman
Janani Suraksha Yojana (JSY)	2005	cash support for institutional (hospital) deliveries, reducing mother and baby deaths
Pulse Polio	1995	mass polio drops campaigns; led to a polio-free India
Ayushman Bharat Health & Wellness Centres	2018	upgrade primary centres for preventive and basic care near homes

PYQ PATTERN: Launch years of PM-JAY (2018) and Mission Indradhanush (2014) and polio-free year (2014) are standard one-liners.

Health in Rajasthan - State Efforts

- Rajasthan runs its public hospitals, community health centres and primary health centres through the state health department, strengthened under the National Health Mission with **ASHA workers** connecting villages to the system.
- The **Mukhyamantri Chiranjeevi Swasthya Bima Yojana** (launched 2021) gives cashless treatment cover to families in empanelled hospitals — Rajasthan's own big step towards universal health coverage, running alongside PM-JAY ideas.
- **Anaemia and malnutrition** are Rajasthan's stubborn health challenges, addressed through iron-folic acid supplementation, midday meals and anganwadi nutrition under **POSHAN Abhiyaan** (the national nutrition mission, launched 2018).
- **ORS (Oral Rehydration Solution)** — a simple mix of salt, sugar and clean water — remains the cheapest life-saving technology in diarrhoeal disease, used from village homes to big hospitals.

MEMORY HOOK: For the Rajasthan angle, remember the trio: Chiranjeevi (insurance), ASHA (village link), ORS (three-ingredient life-saver).

Environment, Biodiversity & Conservation

Everything alive is connected: plants make food, animals eat, microbes recycle. When this web breaks, humans suffer first. RAS tests this chapter with definitions, hotspots, acts and years — all learnable as clean lists.

Ecosystem Basics

- An **ecosystem** is a community of living beings plus their non-living surroundings (air, water, soil) working together. **Producers** (green plants) make food, **consumers** eat, and **decomposers** (bacteria, fungi) recycle dead matter.

- Food chains link into **food webs**. Energy flows one way and decreases at each level — only about 10% passes up (the "10% law").
- A **biome** is a very large ecosystem type: desert (Thar), grassland, rainforest, tundra.

Environmental Change and Impact Assessment

Greenhouse gases like carbon dioxide and methane trap heat, driving **climate change**: hotter summers, erratic monsoons, glacier melt and sea-level rise. Understanding likely harm **before** starting a project is the job of **EIA (Environmental Impact Assessment)** — a study that predicts a project's effects on air, water, land, wildlife and people, and plans how to reduce them. In India, many projects (mines, dams, industries) legally need an environmental clearance based on EIA before they start.

PYQ PATTERN: "EIA full form and purpose" is asked as a direct question — Environmental Impact Assessment, done before project approval.

Biodiversity

Biodiversity means the variety of life at three levels: **genetic diversity** (within a species), **species diversity** (number of species), and **ecosystem diversity** (variety of habitats).

- **Biodiversity hotspots** are regions very rich in unique species but heavily threatened. A hotspot must have at least 1,500 endemic plant species and have lost most of its original habitat. The world has over 35 hotspots.
- India's four hotspots: **Himalaya**, **Indo-Burma**, **Western Ghats & Sri Lanka**, and **Sundaland** (Nicobar Islands part).

MEMORY HOOK: India's hotspot four — Himalaya, Indo-Burma, Western Ghats-Sri Lanka, Sundaland. Three are mountain/island belts; Sundaland enters India only through the Nicobars.

- The **IUCN Red List** classifies species by extinction risk: Least Concern → Near Threatened → Vulnerable → Endangered → Critically Endangered → Extinct in the Wild → Extinct. The Asiatic lion is Endangered; the Great Indian Bustard (Rajasthan's state bird) is **Critically Endangered**.

Conservation - In-situ and Ex-situ

Type	Meaning	Examples
In-situ (on site)	protecting species in their natural home	national parks, wildlife sanctuaries, biosphere reserves
Ex-situ (off site)	protecting outside the natural home	zoos, botanical gardens, seed banks, gene banks

- **National parks** are strictly protected (no grazing/habitat interference); **sanctuaries** allow some regulated human activity. Rajasthan examples: **Ranthambore** and **Sariska** (tiger reserves), **Keoladeo Ghana, Bharatpur** (famous bird sanctuary, a World Heritage Site), **Desert National Park** (home of the Great Indian Bustard).
- **Project Tiger** was launched in **1973** (first at Corbett) to save the Bengal tiger; today India holds the majority of the world's wild tigers. **Project Elephant** followed in 1992.
- **Biosphere reserves** combine a protected core with zones where people live sustainably; India is part of UNESCO's world network.

Key Acts

Act	Year	Purpose
Indian Forest Act legacy / Forest Conservation Act	1927 / 1980	control over forests; 1980 Act checks diversion of forest land
Wildlife (Protection) Act	1972	legal protection to wild animals, birds and plants; created protected areas
Environment (Protection) Act	1986	umbrella law after the Bhopal gas tragedy; powers to set standards and punish polluters
Biological Diversity Act	2002	conservation and fair sharing of benefits from biological resources

EXAM TRAP: Environment Protection Act is **1986**, not 1972 — 1972 is Wildlife Protection. This year-swap is a certified RAS trap.

Global Agreements You Must Know

Agreement	Year	What it does
Montreal Protocol	1987	phases out ozone-destroying substances (CFCs); the ozone layer is recovering
UNFCCC (Earth Summit, Rio)	1992	the framework treaty under which climate talks happen
Kyoto Protocol	1997	set emission-cut targets for developed countries
Paris Agreement	2015	all countries pledged to keep global warming well below 2 degree C, aiming for 1.5 degree C
Ramsar Convention	1971	protects wetlands of international importance; Rajasthan's Sambhar Lake and Keoladeo are Ramsar sites
Convention on Biological Diversity	1992	global treaty for conserving biodiversity, signed at Rio

- India has committed to reach **net zero emissions by 2070** (announced at COP26, Glasgow, 2021) and to expand non-fossil electricity capacity — both are current-affairs-linked facts RAS may touch.

EXAM TRAP: Montreal = ozone (1987), Paris = climate (2015). Examiners cross these two more than any other pair.

Natural Resources and Sustainable Development

Resources like forests, water and minerals are capital, not income — spending them all leaves nothing for the next generation. **Sustainable development**, defined by the Brundtland Report (1987), means "development that meets the needs of the present without compromising the ability of future generations to meet their own needs." The UN's **17 Sustainable Development Goals (SDGs, 2015, target year 2030)** give the global checklist — clean water, climate action, life on land, zero hunger and more. India's own traditions — water harvesting (johads of Rajasthan), sacred groves, khejri protection by the Bishnoi community — are living examples of this idea, centuries before the term existed.

PYQ PATTERN: The Bishnoi sacrifice of 1730 at Khejarli (Amrita Devi Bishnoi and 362 others protecting khejri trees) is a beloved Rajasthan fact — it links environment, Rajasthan GK and values in one question.

Agriculture, Horticulture, Forestry & Animal Husbandry - Rajasthan Focus

Rajasthan is India's largest state and one of its driest: most of it is the Thar desert or semi-arid land. Farming science here is the science of surviving with little water — right crops, right trees, right animals. RAS gives this chapter a strong Rajasthan flavour.

Crop Seasons and the Green Revolution

India's farm year has three seasons, and Rajasthan follows the same rhythm with its own crop mix.

Season	Sowing - Harvest	Main crops
Kharif	sown June-July with monsoon, harvested Sept-Oct	bajra, jowar, maize, rice, cotton, groundnut, moong, moth
Rabi	sown Oct-Nov in winter, harvested March-April	wheat, mustard, gram, barley, cumin
Zaid	short summer season between rabi and kharif	watermelon, muskmelon, fodder crops, vegetables

- Rajasthan is India's largest producer of **mustard** among the major rabi oilseeds and leads in bajra, guar and several seed spices — facts that combine agriculture with state GK.
- The **Green Revolution** of the 1960s-70s combined high-yielding variety (HYV) seeds, chemical fertilisers, irrigation and pesticides, led scientifically by **M.S. Swaminathan** in India. It made India self-sufficient in foodgrains but also raised concerns: falling water tables, soil fatigue and fertiliser overuse — which is why today's policy talks of an **Evergreen Revolution** (productivity without ecological harm) and of millets, promoted globally in the **International Year of Millets (2023)**.
- **Soil health cards** give farmers laboratory-based advice on which nutrients their field actually needs, replacing guesswork fertiliser use.

EXAM TRAP: Mustard and wheat are **rabi**; bajra and cotton are **kharif**. Season-swapping is the simplest and most frequent agriculture trap.

Arid-Zone Agriculture

- Agriculture in dry zones depends on **drought-resistant, short-duration crops** that finish before stored soil moisture runs out.
- **Bajra (pearl millet)** is the king crop of arid Rajasthan — it tolerates heat, poor sandy soil and low rainfall, and gives both grain and fodder.
- **Moth bean** is among the most drought-hardy pulses known — it survives where almost nothing else grows, and its spreading plant cover protects soil from wind erosion.
- Other hardy crops: moong, guar (cluster bean — also an industrial gum source), sesame (til) and castor.
- Water wisdom: **khadin** farming (runoff harvesting, perfected by the Paliwal Brahmins of Jaisalmer), drip and sprinkler irrigation, and drought-tolerant varieties from research centres like **CAZRI (Central Arid Zone Research Institute, Jodhpur)**.

MEMORY HOOK: In the desert, the best technology is choosing the crop that asks for the least water — bajra first, moth always.

Khejri - The Lifeline Tree

- **Khejri** (Prosopis cineraria) is the **state tree of Rajasthan** and is called the **Kalpavriksha of the Thar**. Its deep roots reach groundwater; its leaves (loong) feed animals, its pods (sangri) are a prized vegetable of Marwar cuisine, and it enriches soil by fixing nitrogen.
- Farmers deliberately keep khejri standing in cropped fields — a traditional agroforestry system older than the textbook term.
- The **Bishnoi community** protects khejri and blackbuck as sacred; the 1730 Khejarli sacrifice remains the moral foundation of desert conservation.

Horticulture in Dry Lands

- **Ber (Indian jujube)** fruits well in arid areas with little irrigation and is called the "poor man's apple" of the desert.
- **Pomegranate (anar)** has expanded in Jalore-Barmer belts with drip irrigation; it rewards careful water management with high market value.
- **Date palm (khajoor)** suits the hottest, driest pockets like Jaisalmer and Bikaner; tissue-culture plants improved its establishment.
- Isabgol, cumin, coriander and other seed spices also make Rajasthan a leader in low-water, high-value farming.

Forestry in Rajasthan

- Rajasthan's forests are mostly **dry deciduous and thorn forests**: dhok (Anogeissus) on the Aravalli slopes, teak in the southern districts (Banswara-Dungarpur belt), salar, khair, bamboo in wetter pockets, and khejri-babool scrub in the desert.
- **Social forestry** means planting trees on community land, roadsides, canal banks and farms to meet village needs for fuel, fodder and small timber.
- **Joint Forest Management (JFM)** involves village committees in protecting and sharing benefits from nearby forests — protection works when local people gain from it.

Animal Husbandry and Breeds

Animal husbandry is the desert economy's backbone — livestock gives milk, income and insurance in drought years. Rajasthan is among India's top milk-producing states.

Animal	Rajasthan-linked breeds	Breed note
Cattle	Tharparkar, Rathi, Gir, Nagori	Tharparkar from the Thar region — hardy dual-purpose; Rathi from Bikaner belt, a good milker; Nagori famous as draught animals
Buffalo	Murrah, Surti	Murrah is India's best-known high-yield buffalo (Haryana-Rajasthan tract); Surti is smaller and hardy
Sheep	Marwari, Chokla	Marwari — hardy desert sheep; Chokla gives finer wool, called a quality wool breed

Animal	Rajasthan-linked breeds	Breed note
Goat	Sirohi, Jakhrana	Sirohi is the major meat-milk goat of the Aravalli region; Jakhrana from Alwar side
Camel	Bikaneri, Jaisalmeri	Bikaneri is the main riding/draught type; the camel is Rajasthan's **state animal** (in the livestock category) — Pushkar fair is world-famous

- **Artificial Insemination (AI)** uses semen from superior bulls to improve whole herds quickly — the main engine of breed improvement, along with cross-breeding and veterinary care.
- Common animal diseases decide rural incomes: see the table below.

Disease	Cause	Key point
Foot-and-Mouth Disease (FMD)	virus	highly infectious; blisters in mouth and feet; controlled by regular vaccination
Anthrax	bacterium (Bacillus anthracis)	affects animals and can spread to humans; carcass disposal is critical
Haemorrhagic Septicaemia (HS)	bacterium	sudden monsoon-season killer of cattle and buffalo
Brucellosis	bacterium	causes abortion in animals; also infects humans

EXAM TRAP: FMD is **viral**, anthrax is **bacterial** — a very common swapped-option trap.

White Revolution and Operation Flood

- **Operation Flood (1970-1996)**, run by the **NDDB (National Dairy Development Board)**, made India the **world's largest milk producer**. Its architect was **Dr Verghese Kurien**, the "Father of the White Revolution", who built the Anand cooperative model of Gujarat (Amul) and replicated it nationally through village cooperatives.
- The cooperative pattern — farmers own the village society, the district union processes, the state federation markets — reached Rajasthan too, where the state federation markets milk under the **Saras** brand.
- Milk is called a nearly complete food: it carries protein, fat, calcium and vitamins together, which is why dairying transforms nutrition as well as income.

PYQ PATTERN: Kurien + NDDB + Operation Flood + "world's largest milk producer" is a near-certain one-liner cluster. Add "Saras" for the Rajasthan angle.

Indian Contributions, Programmes & Indigenisation

India's science story runs from ancient mathematics to quantum missions. RAS loves the "who did what" table and the recent mission names — both are gift marks if revised once, properly.

Great Indian Scientists

Scientist	Contribution	Memory line
C.V. Raman	Raman Effect (scattering of light) — Nobel Prize 1930	**National Science Day, 28 February**, marks the day of the discovery (1928)
Homi J. Bhabha	father of India's atomic energy programme	founded TIFR (1945) and led the Atomic Energy Commission
Vikram Sarabhai	father of the Indian space programme	founded ISRO's path; also IIM Ahmedabad
A.P.J. Abdul Kalam	Missile Man of India; led IGMDP	later President of India (2002-2007)
J.C. Bose	pioneer of radio/wireless work and plant physiology	showed plants respond to stimuli; invented crescograph
Srinivasa Ramanujan	self-taught mathematics genius — number theory	his birthday, 22 December, is National Mathematics Day
Har Gobind Khorana	helped crack the genetic code — Nobel Prize 1968	also built a gene artificially in the laboratory
S. Chandrasekhar	Chandrasekhar limit for white dwarf stars — Nobel Prize 1983	calculated the mass limit (about 1.4 solar masses)
M.S. Swaminathan	father of India's Green Revolution	introduced high-yielding wheat varieties; agronomist-statesman
Satyendra Nath Bose	Bose-Einstein statistics with Einstein	the "boson" particle family is named after him
Meghnad Saha	Saha ionisation equation in astrophysics	also a founder of science planning in India
Birbal Sahni	father of Indian palaeobotany (fossil plants)	founded the institute at Lucknow named after him

MEMORY HOOK: Two February-December anchors: Science Day 28 Feb (Raman), Mathematics Day 22 Dec (Ramanujan). Examiners swap them — don't let them.

EXAM TRAP: Ramanujan = Mathematics Day (22 Dec). Sarabhai = space. Bhabha = atomic energy. Kalam = missiles. Four "father of" titles, four different men.

Recent National S&T Programmes and Missions

Mission / programme	Field	Purpose in one line
National Quantum Mission	quantum technology	build quantum computers, communication and sensing capacity in India
National Green Hydrogen Mission	clean energy	make India a global hub for green hydrogen production and export
India Semiconductor Mission	chips	support chip design, fabrication and packaging units in India
Deep Ocean Mission	oceans	explore deep-sea minerals and life; includes the Samudrayaan manned submersible (Matsya 6000)
Vigyan Dhara	science funding	a central scheme that streamlines science and technology funding to institutions and researchers

Mission / programme	Field	Purpose in one line
National Supercomputing Mission	computing	build a grid of indigenous supercomputers across research institutions
Anusandhan National Research Foundation (ANRF)	research funding	national body to fund and mentor research in universities and institutions

- These missions share one pattern: reduce import dependence in the technologies that will decide the future — chips, hydrogen, quantum and deep-sea resources.

Science & Technology in Rajasthan

- The **Department of Science & Technology, Government of Rajasthan** promotes science popularisation, research support and innovation in the state.
- **Regional science centres and science parks** (the Regional Science Centre at Jaipur and district-level science activities) take hands-on science to school students.
- State research strength: **CAZRI** and the Arid Forest Research Institute (Jodhpur) lead desert science; **MPUAT (Udaipur)** and the agriculture universities drive farm research; Rajasthan also hosts a campus of national institutes and the Birla science institutions at Pilani — **BITS Pilani** grew from the Birla education trust and remains a landmark of technical education in the state.
- Rajasthan applies S&T through schemes like solar pumps for farmers, remote-sensing based water and land planning, and digital service delivery through e-Mitra.

Indigenisation of Technology

Indigenisation means designing and building technology at home instead of importing it — it saves foreign exchange, protects security, and keeps knowledge in the country.

- **Tejas LCA**: an Indian-designed fighter aircraft replacing imported fleets step by step.
- **Kaveri engine**: India's long effort to build a jet engine for fighters — it taught hard lessons and its technology now feeds drone and future engine programmes; engine technology is among the toughest to indigenise.
- **BrahMos**: started as a joint venture and is steadily raising Indian content — missiles, launchers and systems increasingly made in India.
- **UPI**: a fully Indian-built digital payment system, now processing billions of transactions monthly and being adopted or linked by other countries — proof that indigenisation is not only about weapons; it includes public digital platforms like Aadhaar-linked services, DigiLocker and CoWIN.
- Space tells the same story: the indigenous **cryogenic engine** turned denial into capability, and ISRO now launches foreign satellites commercially through its commercial arms.

PYQ PATTERN: "Indigenisation" questions usually give four examples and ask which is/are correct — Tejas (aircraft), BrahMos (missile, JV with rising Indian share), Kaveri (engine programme), UPI (digital platform). All four are valid examples.

MEMORY HOOK: Close the chapter with one line — Indian science's byline is simple: denied yesterday, designed today. If you can attach one name, one year and one example to each row of these tables, this whole unit is yours.